

Nama : Alberta Aurora Adiswari

NIM : 2311102220

Kelas : S1IF-11-REG03

MODUL 12 dan 13 LARAVEL:Database 1 dan 2

1. Laravel Database 1 Model

```
<?php

namespace App\Models;

use Illuminate\Contracts\Auth\MustVerifyEmail;
use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Foundation\Auth\User as Authenticatable;
use Illuminate\Notifications\Notifiable;
use Laravel\Sanctum\HasApiTokens;

class User extends Authenticatable
{
    use HasApiTokens, HasFactory, Notifiable;

    /**
     * The attributes that are mass assignable.
     *
     * @var array<int, string>
     */
    protected $fillable = [
        'name',
        'email',
        'password',
    ];

    /**
     * The attributes that should be hidden for serialization.
     *
     * @var array<int, string>
     */
    protected $hidden = [
        'password',
        'remember_token',
    ];

    /**
     * The attributes that should be cast.
     *
     * @var array<string, string>
     */
    protected $casts = [
        'email_verified_at' => 'datetime',
    ];
}
```

```
<?php
```

```

namespace App\Models;

use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Database\Eloquent\Model;

class Product extends Model
{
    use HasFactory;

    protected $fillable = [
        'name',
        'price',
    ];

    public function variants()
    {
        return $this->hasMany(Variant::class);
    }
}

```

```

<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Database\Eloquent\Model;

class Variant extends Model
{
    use HasFactory;

    protected $fillable = [
        'product_id',
        'name',
        'price',
    ];

    public function product()
    {
        return $this->belongsTo(Product::class);
    }
}

```

Controller

```

<?php

namespace App\Http\Controllers;

use Illuminate\Foundation\Auth\Access\AuthorizesRequests;
use Illuminate\Foundation\Bus\DispatchesJobs;
use Illuminate\Foundation\Validation\ValidatesRequests;
use Illuminate\Routing\Controller as BaseController;

class Controller extends BaseController
{
    use AuthorizesRequests, DispatchesJobs, ValidatesRequests;
}

```

```

<?php

namespace App\Http\Controllers;

use App\Models\Product;
use Illuminate\Http\Request;

class ProductController extends Controller
{
    public function index()
    {
        $products = Product::all();

        return view('products.index', compact('products'));
    }

    public function create()
    {
        return view('products.form', [
            'title' => 'Tambah',
            'product' => new Product(),
            'route' => route('products.store'),
            'method' => 'POST',
        ]);
    }

    public function store(Request $request)
    {
        $validated = $request->validate([
            'name' => 'required|min:4',
            'price' => 'required|integer|min:1000000',
        ]);

        Product::create($validated);

        return redirect()->route('products.index')
            ->with('success', 'Produk berhasil ditambahkan');
    }

    public function show(Product $product)
    {
        $product->load('variants');

        return view('products.show', compact('product'));
    }

    public function edit(Product $product)
    {
        return view('products.form', [
            'title' => 'Edit',
            'product' => $product,
            'route' => route('products.update', $product),
            'method' => 'PUT',
        ]);
    }

    public function update(Request $request, Product $product)
    {

```

```

        $validated = $request->validate([
            'name' => 'required|min:4',
            'price' => 'required|integer|min:1000000',
        ]);

        $product->update($validated);

        return redirect()->route('products.index')
            ->with('success', 'Produk berhasil diperbarui');
    }

    public function destroy(Product $product)
    {
        $product->delete();

        return redirect()->route('products.index')
            ->with('success', 'Produk berhasil dihapus');
    }
}

```

```

<?php

namespace App\Http\Controllers;

use Illuminate\Http\Request;
use Illuminate\Support\Facades\Auth;

class SiteController extends Controller
{
    public function auth(Request $request)
    {
        $credentials = $request->validate([
            'email' => 'required|email',
            'password' => 'required',
        ]);

        if (Auth::attempt($credentials)) {
            $request->session()->regenerate();

            return redirect()->route('products.index')
                ->with('success', 'Login berhasil. Selamat datang, ' . Auth::user()->name);
        }

        return redirect('/login')
            ->with('msg', 'Email / password salah')
            ->withInput();
    }

    public function logout(Request $request)
    {
        Auth::logout();

        $request->session()->invalidate();
        $request->session()->regenerateToken();

        return redirect('/login')->with('msg', 'Berhasil logout');
    }
}

```

View

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Login</title>

  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">

  <script src="{{ asset('js/jquery.min.js') }}"></script>
</head>
<body class="bg-light">

  <div class="container">
    <div class="row justify-content-center align-items-center min-vh-100">
      <div class="col-md-4">

        <div class="card">
          <div class="card-body">
            <h3 class="card-title text-center mb-4">Login</h3>

            <form method="POST" action="{{ route('auth') }}">
              @csrf

              <div class="mb-3">
                <label for="email" class="form-label">Email</label>
                <input type="email" name="email" id="email" class="form-control" required>
              </div>

              <div class="mb-3">
                <label for="password" class="form-label">Password</label>
                <input type="password" name="password" id="password" class="form-
control" required>
              </div>

              <div class="d-grid">
                <button type="submit" class="btn btn-primary">Login</button>
              </div>
            </form>

          </div>
        </div>

      </div>
    </div>
  </div>

  <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

```
<!DOCTYPE html>
<html>
<head>
```

```

<title>Daftar Produk</title>
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
</head>
<body class="container mt-5">

  <div class="d-flex justify-content-end mb-3">
    <span class="me-3">Login sebagai: <strong>{{ auth()->user()->name }}</strong></span>
    <a href="{{ route('logout') }}" class="btn btn-warning btn-sm">Logout</a>
  </div>

  <h3>Daftar Produk</h3>

  @if(session('success'))
    <div class="alert alert-success">
      {{ session('success') }}
    </div>
  @endif

  <a href="{{ route('products.create') }}" class="btn btn-primary mb-3">Tambah Produk</a>

  <table class="table table-bordered">
    <tr>
      <th>No</th>
      <th>Nama Produk</th>
      <th>Harga</th>
      <th>Aksi</th>
    </tr>

    @forelse($products as $product)
      <tr>
        <td>{{ $loop->iteration }}</td>
        <td>{{ $product->name }}</td>
        <td>Rp {{ number_format($product->price, 0, ',', '.') }}</td>
        <td>
          <a href="{{ route('products.show', $product->id) }}" class="btn btn-info btn-sm">Detail</a>

          <a href="{{ route('products.edit', $product->id) }}" class="btn btn-warning btn-sm">Edit</a>

          <form action="{{ route('products.destroy', $product->id) }}" method="POST"
style="display:inline;">
            @csrf
            @method('DELETE')

            <button type="submit" class="btn btn-danger btn-sm" onclick="return
confirm('Yakin hapus data?')">
              Hapus
            </button>
          </form>
        </td>
      </tr>
    @empty
      <tr>
        <td colspan="4" class="text-center">Belum ada data produk</td>
      </tr>
    @endforelse
  </table>

```

```
</body>
</html>
```

```
<!DOCTYPE html>
<html>
<head>
  <title>{{ $title }} Produk</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
</head>
<body class="container mt-5">

  <h3>Form {{ $title }} Produk</h3>

  <form action="{{ $route }}" method="POST">
    @csrf

    @if($method == 'PUT')
      @method('PUT')
    @endif

    <div class="mb-3">
      <label>Nama Produk</label>
      <input type="text" name="name" class="form-control @error('name') is-invalid @enderror"
        value="{{ old('name', $product->name) }}">

      @error('name')
        <div class="invalid-feedback">{{ $message }}</div>
      @enderror
    </div>

    <div class="mb-3">
      <label>Harga</label>
      <input type="number" name="price" class="form-control @error('price') is-invalid
@enderror"
        value="{{ old('price', $product->price) }}">

      @error('price')
        <div class="invalid-feedback">{{ $message }}</div>
      @enderror
    </div>

    <button type="submit" class="btn btn-success">Simpan</button>
    <a href="{{ route('products.index') }}" class="btn btn-secondary">Kembali</a>
  </form>

</body>
</html>
```

```
<!DOCTYPE html>
<html>
<head>
  <title>Detail Produk</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
</head>
<body class="container mt-5">
```

```

<h3>Detail Produk</h3>

<div class="card mb-3">
  <div class="card-body">
    <p><strong>Nama Produk:</strong> {{ $product->name }}</p>
    <p><strong>Harga:</strong> Rp {{ number_format($product->price, 0, ',', '.') }}</p>
  </div>
</div>

<h4>Daftar Variant</h4>

<table class="table table-bordered">
  <tr>
    <th>No</th>
    <th>Nama Variant</th>
    <th>Harga Variant</th>
  </tr>

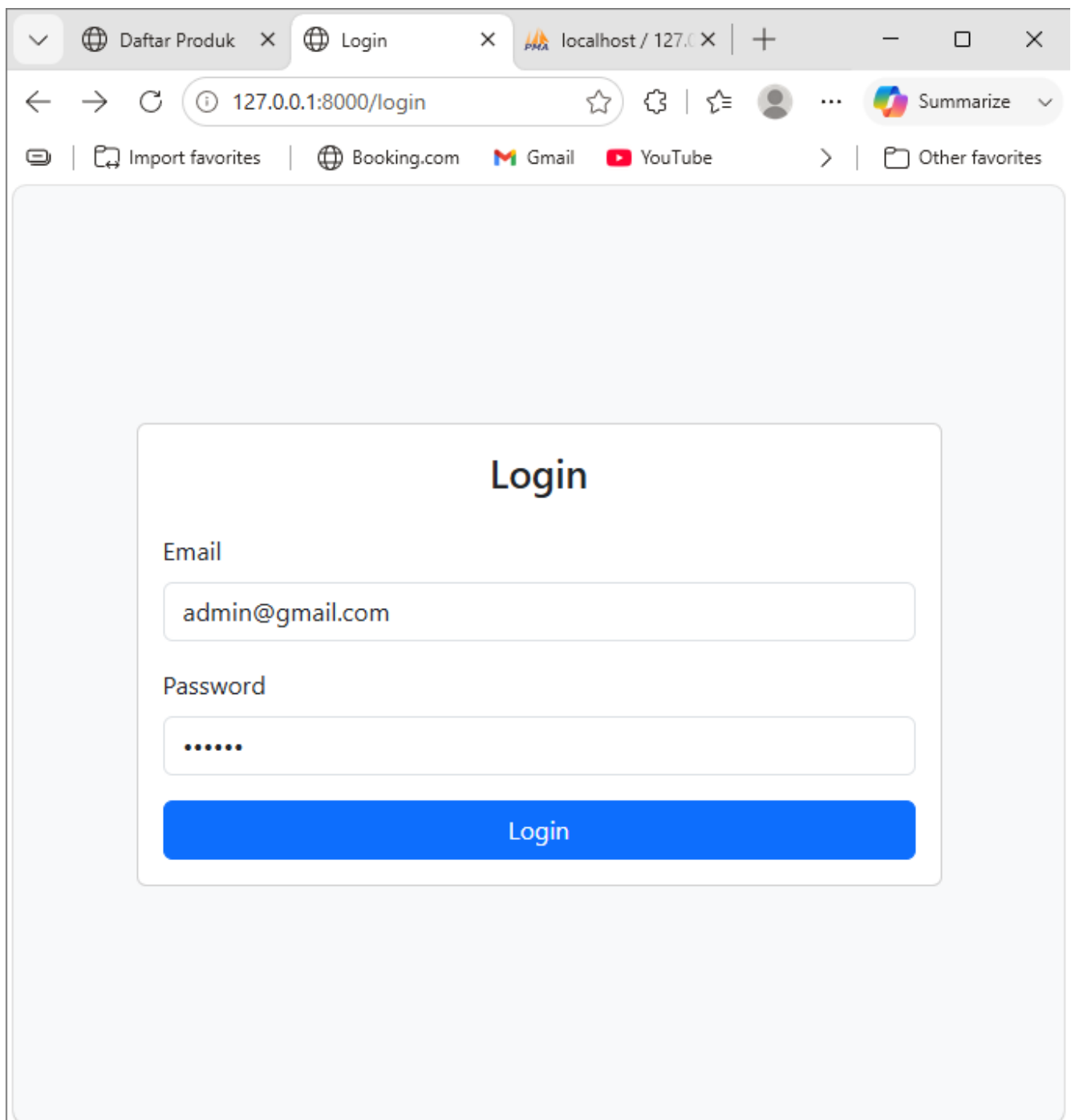
  @forelse($product->variants as $variant)
    <tr>
      <td>{{ $loop->iteration }}</td>
      <td>{{ $variant->name }}</td>
      <td>Rp {{ number_format($variant->price, 0, ',', '.') }}</td>
    </tr>
  @empty
    <tr>
      <td colspan="3" class="text-center">Produk ini belum memiliki variant</td>
    </tr>
  @endforelse
</table>

<a href="{{ route('products.index') }}" class="btn btn-secondary">Kembali</a>

</body>
</html>

```

Screenshoot program



Daftar Produk

Daftar Produk

localhost / 127.0.0.1:8000/products

127.0.0.1:8000/products

Summarize

Import favorites

Booking.com

Gmail

YouTube

Other favorites

Daftar Produk

Login berhasil. Selamat datang, Admin

Tambah Produk

No	Nama Produk	Harga	Aksi
Belum ada data produk			

Daftar Produk

Daftar Produk

localhost / 127.0.0.1:8000/products

127.0.0.1:8000/products

Summarize

Import favorites

Booking.com

Gmail

YouTube

Other favorites

Login sebagai: **Admin**

Logout

Daftar Produk

Tambah Produk

No	Nama Produk	Harga	Aksi
Belum ada data produk			

Daftar Produk

Daftar Produk

localhost / 127.0.0.1:8000/products

127.0.0.1:8000/products

Import favorites | Booking.com | Gmail | YouTube | Other favorites

Login sebagai: Admin

Logout

Daftar Produk

Login berhasil. Selamat datang, Admin

Tambah Produk

No	Nama Produk	Harga	Aksi
1	Kertas HVS	Rp 1.000.000	DetailEditHapus

127.0.0.1:8000/products/2

Daftar ProdukDetail Produklocalhost / 127.0.0.1 / ecommerce

localhost/phpmyadmin/index.php?route=/database/structure&db=ecommerce

Import favoritesBooking.comGmailYouTubeMapsUtomikOther favorites

phpMyAdmin

RecentFavorites

dbshop

ecommerce

New

failed_jobs

migrations

password_resets

personal_access_tokens

products

users

variants

information_schema

kursus_web

mysql

performance_schema

petclub

phpmyadmin

sasana_educasi

test

we

webpro_login

weddingdb

Server: 127.0.0.1Database: ecommerce

StructureSQLSearchQueryExportImportOperationsPrivilegesRoutinesEventsTriggers

Filters

Containing the word:

Table	Action	Rows	Type	Collation	Size	Overhead
☐ failed_jobs		0	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
☐ migrations		6	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
☐ password_resets		0	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
☐ personal_access_tokens		0	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
☐ products		1	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
☐ users		1	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
☐ variants		2	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
7 tables	Sum	10	InnoDB	utf8mb4_general_ci	208.0 KiB	0 B

☐ Check allWith selected:

PrintData dictionary

Create new table

ConsoleheNumber of columns

Server: 127.0.0.1Database: ecommerceTable: users

BrowseStructureSQLSearchInsertExportImportPrivilegesOperationsTrackingTriggers

Showing rows 0 - 0 (1 total, Query took 0.0004 seconds.)

SELECT * FROM `users`

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show allNumber of rows: 25Filter rows: Search this table

Extra options

Edit

Copy

Delete

1 Adminadmin@gmail.comNULL\$2y\$10\$PdFlyGRTIMXDheAI70zi.yLEQOIFhmiaYA6.rF5sw9...NULL2026-0604:03:34

☐ Check allWith selected: EditCopyDeleteExport

☐ Show allNumber of rows: 25Filter rows: Search this table

Query results operations

Print

Copy to clipboard

Export

Display chart

Create view

Consolemark this SQL query

Server: 127.0.0.1 » Database: ecommerce » Table: variants

Showing rows 0 - 1 (2 total, Query took 0.0004 seconds.)

`SELECT * FROM `variants``

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	id	product_id	name	price	created_at	updated_at
☐ Edit Copy Delete	1	2	Variant Merah	1200000	2026-06-23 04:26:05	2026-06-23 04:26:05
☐ Edit Copy Delete	2	2	Variant Hitam	1500000	2026-06-23 04:26:05	2026-06-23 04:26:05

Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

Console mark this SQL query

Penjelasan kode dan output

Pada praktikum Modul 12 dan 13, aplikasi Laravel dikembangkan menggunakan database MySQL dengan nama database ecommerce. Konfigurasi database dilakukan melalui file .env, kemudian migration dijalankan untuk membuat tabel products dan variants. Tabel products digunakan untuk menyimpan data produk, sedangkan tabel variants digunakan untuk menyimpan data varian produk yang terhubung melalui product_id. Model Product dan Variant digunakan untuk menghubungkan aplikasi dengan database serta membentuk relasi one to many, yaitu satu produk dapat memiliki banyak variant.

Kode pada ProductController digunakan untuk menjalankan proses CRUD produk, meliputi menampilkan data, menambah, mengedit, menghapus, dan melihat detail produk. Validasi form diterapkan agar nama produk wajib diisi minimal 4 karakter dan harga minimal 1.000.000. Pada Modul 13, fitur login, logout, session, dan middleware ditambahkan melalui SiteController dan route products diberi middleware auth agar hanya user yang sudah login dapat mengakses halaman produk. Output menunjukkan aplikasi berhasil menampilkan halaman login, daftar produk, form tambah/edit, pesan validasi, serta detail produk beserta variant-nya, sehingga fitur CRUD, validasi, autentikasi, middleware, dan relasi model berhasil dijalankan.